
A STUDY OF SPATIAL PYRAMID ATTENTION NETWORKS FOR IMAGE MANIPULATION LOCALISATION

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Contents

Abstract	1
Acknowledgments	2
1 Introduction	3
2 SPAN Architecture	7
2.1 Feature Extractor	8
2.1.1 Feature Adaptation	8
2.2 Pyramid Spatial Attention	9
2.2.1 Positional Projection	10
2.2.2 Residual Connection	10
2.2.3 Pyramid Propagation	11
2.2.4 Why 3×3 Blocks?	12
2.3 Decision Module	13
2.4 Training Objective	14
3 Experiments & Performance Analysis	15
3.1 Experimental Setup	15
3.1.1 Datasets	15
3.1.2 Models Under Evaluation	16
3.1.3 Robustness Perturbation Conditions	16
3.1.4 Evaluation Protocol	17
3.2 Results	18
3.2.1 Columbia Dataset — Pixel-Level AUC	18
3.2.2 Coverage Dataset — Pixel-Level AUC and F1	19
3.2.3 CASIA v1 Dataset — Image-Level Evaluation	20
3.3 Overall Analysis and Discussion	20
3.3.1 Robustness Trends	20
3.3.2 SPAN vs. ManTraNet	21
3.3.3 Limitations and Resource Constraints	21
3.4 Conclusion	21
References	22

Abstract

This report presents a study and partial replication of the Spatial Pyramid Attention Network (SPAN), a deep learning framework for detecting and localising manipulated regions in images. The primary focus is on developing a thorough understanding of the SPAN architecture and reproducing its core experimental results. SPAN models relationships between image patches at multiple scales using a hierarchy of local self-attention blocks, producing pixel-wise predictions of which regions in an image have been tampered with. The original TensorFlow implementations of both SPAN and a baseline method, ManTraNet, were re-implemented in PyTorch using pre-trained weights and evaluated without any fine-tuning across multiple benchmark datasets under a range of perturbation conditions to assess robustness. SPAN consistently and substantially outperforms ManTraNet across all settings, demonstrating strong generalisation to multiple manipulation types without task-specific retraining.

Acknowledgments

Huge shoutout to Kavita for teaching the Computer Vision course with remarkable zeal and thoroughness — and for adding one more lens to my perception. The classes were highly interactive, the breadth of papers we studied was engaging, and I developed both an interest in the subject and the habit of reading research papers, which naturally paved the way for this study.

This project was an exercise in understanding carefully and implementing faithfully. This topic, unlike most well-trodden deep learning models, offered little comfort from the usual sources — no YT walkthroughs, slides, or explanatory blogs; the bottomless pit of the internet came up empty. The paper, the architecture, and the mathematics had to be understood on their own terms, which made the process more challenging, but ultimately rewarding.

Special thanks to Eada for lending his high-GPU laptop, helping with the technical heavy lifting, and saving me from paying Google. Thanks also to GitHub Copilot and various LLMs for code assistance, concept clarification, and the occasional debugging miracles. I must end with an unsolicited appreciation for the human visual system — the inspiration that started it all.

Introduction

The digital image has long served as a primary instrument of visual communication — in journalism, scientific publishing, legal proceedings, and everyday social exchange. Yet the same computational tools that enable image creation and sharing also make it trivially easy to alter an image’s content without leaving any trace visible to the human eye. Image manipulation — the process of altering a digital image’s content or appearance by adding, removing, or modifying regions — has quietly grown into one of the most consequential problems in modern digital media.

The breadth of harm is not merely anecdotal. In biomedical research publishing alone, 3.8% of 20,621 papers published across 40 scientific journals between 1995 and 2004 contained problematic figures, with at least half exhibiting features suggestive of deliberate manipulation. Stern et al. estimated that each retracted article could account for a mean of \$392,582 in direct costs — and these figures are now decades outdated. Beyond academia, image forgeries fuel fake news, insurance fraud, blackmail, and political disinformation. The scale and societal cost of image manipulation compel the development of automated tools capable of detecting it reliably and without human supervision.

The most studied manipulation types in the forensics literature are splicing, copy-move forgery, and removal (inpainting). Splicing involves combining content from two or more different source images into a single composite. The forged region carries the noise fingerprint, lighting model, and JPEG compression history of its original source — all of which are statistically inconsistent with the host image, even if invisible to a human observer.



Figure 1.1: Example of forged images. From left to right: pristine donor image and forged image. Source - ManTra-Net paper.

Copy-move forgery duplicates a region from within the same image and pastes it elsewhere, typically to conceal an object or replicate a background element. Because both

the source and destination regions originate from the same image, the statistical inconsistency is subtler than in splicing, making copy-move one of the harder manipulation types to localise. Removal (inpainting) eliminates an object from the scene and fills the resulting void with synthesised background pixels. Modern inpainting methods, including deep network-based approaches, can produce visually seamless results whose only detectable traces are statistical, not perceptual.



Figure 1.2: Illustration of image splicing by combining content from different source images into a single forged image. Source - CASIA v1 dataset.

Beyond these canonical categories, real-world forgeries frequently combine multiple manipulation types in sequence — for example, splicing a face from another image, then applying smoothing and brightness adjustment to blend the boundary — making the detection problem combinatorially harder.

A key assumption underlying all forgery localisation methods is that manipulated regions differ from authentic regions in their colour, intensity, or noise distributions. These inconsistencies arise because a spliced region inherits the camera noise fingerprint of its source device, which differs from that of the host image; JPEG re-compression introduces characteristic DCT quantisation artefacts at block boundaries that are inconsistent across regions with different compression histories; and post-processing operations such as blurring, sharpening, and inpainting leave frequency-domain signatures that trained filters can detect.

Critically, none of these traces are perceptible to a human observer under normal viewing conditions. They exist entirely in the statistical layer of the image — in the residual noise once the image content is subtracted. Detection therefore requires learning what “normal” noise looks like for an authentic image, and identifying pixels whose noise statistics deviate from that expectation.

Prior work on forgery localisation broadly divides into two classes. Type-specific methods — covering splice detection, copy-move detection, removal detection, and enhancement detection — achieve strong performance on their target manipulation type but fail when the manipulation type is unknown or mixed. This limitation is impractical in real-world deployments where a forger need not advertise their technique.

Generalised detection methods attempt to overcome this limitation. RGB-N proposes a two-stream Faster R-CNN that combines RGB features with SRM noise residuals, but its predictions are inherently limited to rectangular bounding boxes, as real forged regions are not rectangular. It also requires fine-tuning on each target dataset to achieve competitive accuracy.

ManTra-Net represents the state of the art in unified, fine-tuning-free localisation prior to SPAN. It jointly trains a VGG-based feature extractor across 385 types of image

manipulation on a large synthetic dataset, then combines it with an LSTM-based local anomaly detection module. ManTra-Net makes pixel-level predictions and achieves competitive results without any domain-specific retraining. However, its localisation module models only the “vertical” relationship of the same spatial location across different feature map scales, without modelling spatial relationships between different image patches. A tampered region’s boundary is defined not merely by how it looks in isolation, but by how it compares to its neighbours. ManTra-Net’s architecture cannot reason about this spatial comparison, which limits its localisation precision.

The Spatial Pyramid Attention Network (SPAN) addresses this structural limitation directly. SPAN models both the vertical (multi-scale) relationships captured by ManTra-Net and the spatial (patch-to-patch) relationships between neighbouring image regions that ManTra-Net omits. The core insight is that a tampered pixel’s noise fingerprint will be inconsistent with its authentic neighbours — and this inconsistency can be detected by explicitly comparing every pixel’s feature vector against its local neighbourhood at multiple scales simultaneously.

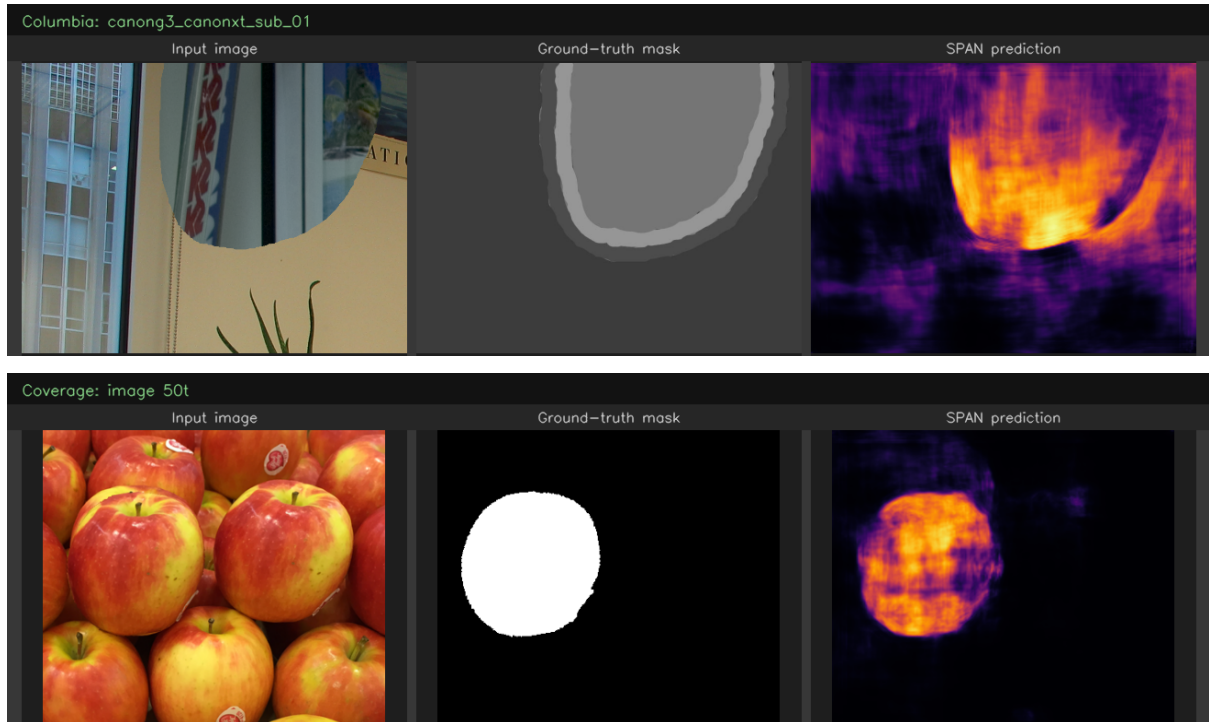


Figure 1.3: Examples of manipulated images involving splicing and copy-move forgery from the Columbia and Coverage datasets, respectively. Left: manipulated image; middle: ground-truth mask; right: SPAN prediction heatmap produced by our implementation using the pretrained weights released by the original authors, without any additional fine-tuning.

SPAN achieves this through a hierarchical local self-attention module that compares each pixel’s learned feature representation against spatial neighbours at exponentially growing dilation distances, forming a spatial pyramid. Authentic pixels exhibit consistent noise across all scales, whereas tampered pixels break the pattern at some scale. This multi-scale spatial reasoning enables SPAN to localise forgery boundaries more precisely

than previous methods without requiring any fine-tuning on the target dataset. Furthermore, SPAN operates using only the potentially manipulated input image and does not require access to the original authentic image for comparison.

SPAN produces a pixel-wise tampering probability map

$$X^{\text{pred}} \in [0, 1]^{H \times W},$$

where values close to 1 indicate high confidence of manipulation and values close to 0 indicate authentic regions. This prediction map is visualised as a heatmap, where darker regions correspond to pixels more likely to be authentic, while brighter yellow regions indicate a higher probability of manipulation. Example shown in Figure 1.3. The binary mask used for quantitative evaluation is obtained by thresholding the prediction map at $\tau = 0.5$.

In this project, a subset of the experiments presented in the original paper was reproduced to evaluate whether a single unified model can detect multiple forms of image manipulation simultaneously, assess its robustness, and compare its performance against ManTra-Net.

SPAN Architecture

When an image is manipulated — by splicing a region from a different photograph or by copy-moving a region within the same image — the manipulation leaves forensic traces. These traces are not visible to the naked eye, but they exist in the statistical structure of the pixel neighbourhood: the camera sensor noise fingerprint changes at the boundary, demosaicing patterns shift, and JPEG compression grids become inconsistent. A forensic system must therefore learn to detect these subtle inconsistencies.

The fundamental challenge is that a tampered pixel does not look suspicious in isolation. It becomes suspicious only when compared against neighbouring regions. A spliced sky region embedded into an indoor photograph may appear visually natural, but its noise statistics differ from those of the authentic indoor background. Traditional CNN-based approaches can detect local anomalies, but cannot explicitly compare spatially distant patches. Conversely, global self-attention architectures can compare all pixels against one another, but incur computational complexity of order $O(N^4)$ for an $N \times N$ image, making them impractical at full image resolution.

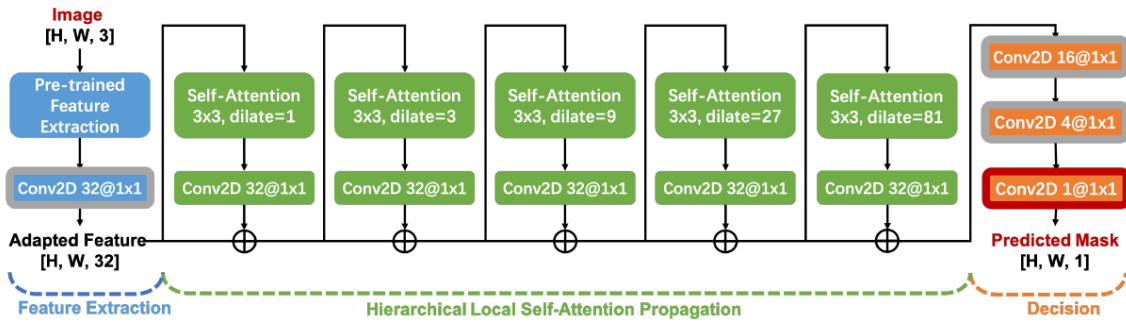


Figure 2.1: Overall architecture of the SPAN framework.

SPAN (Spatial Pyramid Attention Network) addresses this problem by combining local self-attention with hierarchical pyramid propagation. Instead of applying global attention, SPAN uses lightweight local self-attention blocks stacked with exponentially increasing dilation distances. This enables efficient comparison of distant regions while maintaining computational feasibility.

⁰All architecture illustrations in this chapter are taken from the SPAN paper, except Figure 2.2, which is taken from the ManTra-Net paper.

A raw RGB image is first passed through a frozen pre-trained feature extractor inherited from ManTra-Net, producing a 256-channel forensic feature representation. A learnable 1×1 convolution then reduces the channel dimension from 256 to 32 to make self-attention computationally tractable. The resulting feature map is processed through five Local Self-Attention (LSA) blocks with dilation distances $\{1, 3, 9, 27, 81\}$. Finally, a decision module composed of 1×1 convolutions produces a pixel-wise tampering probability map.

2.1 Feature Extractor

The feature extractor used in SPAN is inherited directly from ManTra-Net and remains frozen during SPAN training. The extractor consists of a Wide & Deeper VGG backbone augmented with Bayer convolutional layers and SRM filters.

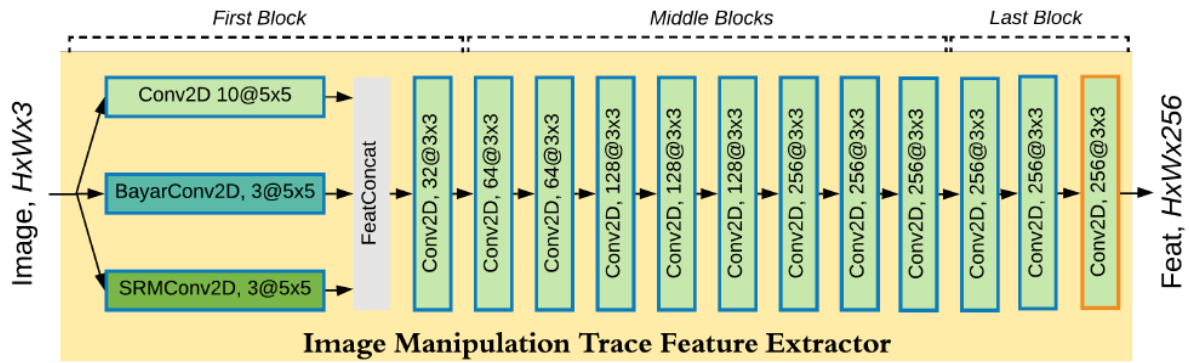


Figure 2.2: Feature extractor inherited from ManTra-Net.

The Bayer convolutional layer is designed to detect inconsistencies in the Color Filter Array (CFA) demosaicing pattern. Since digital cameras reconstruct RGB images from Bayer sensor patterns, manipulated regions originating from different sources often exhibit inconsistent demosaicing statistics near the boundary.

The SRM (Steganalysis Rich Model) filters are fixed high-pass filters that suppress semantic image content and expose residual noise statistics. These residuals capture sensor noise, compression artefacts, and other forensic traces that remain highly consistent in authentic regions but diverge at manipulation boundaries.

The Wide VGG backbone processes these residual representations using stacked 3×3 convolutions to generate a rich forensic embedding:

$$X_{\text{feat}} \in \mathbb{R}^{H \times W \times 256}$$

Importantly, no pooling or spatial downsampling is applied, ensuring that fine-grained forensic traces remain preserved at full image resolution.

2.1.1 Feature Adaptation

Self-attention complexity scales quadratically with feature dimension. Using the full 256-dimensional representation would therefore be computationally prohibitive. To reduce

computational cost, SPAN applies a learnable 1×1 convolution:

$$X = \text{Conv}_{1 \times 1}(X_{\text{feat}}) \in \mathbb{R}^{H \times W \times 32}$$

This operation performs channel-wise dimensionality reduction without altering spatial resolution. The reduction from $256 \rightarrow 32$ decreases attention cost from $256^2 \rightarrow 32^2$ corresponding to a $64\times$ reduction in computational complexity.

2.2 Pyramid Spatial Attention

The central idea of SPAN is that manipulation detection requires comparison between neighbouring image regions rather than isolated pixel analysis. To achieve this efficiently, SPAN uses Local Self-Attention (LSA) blocks arranged in a spatial pyramid.

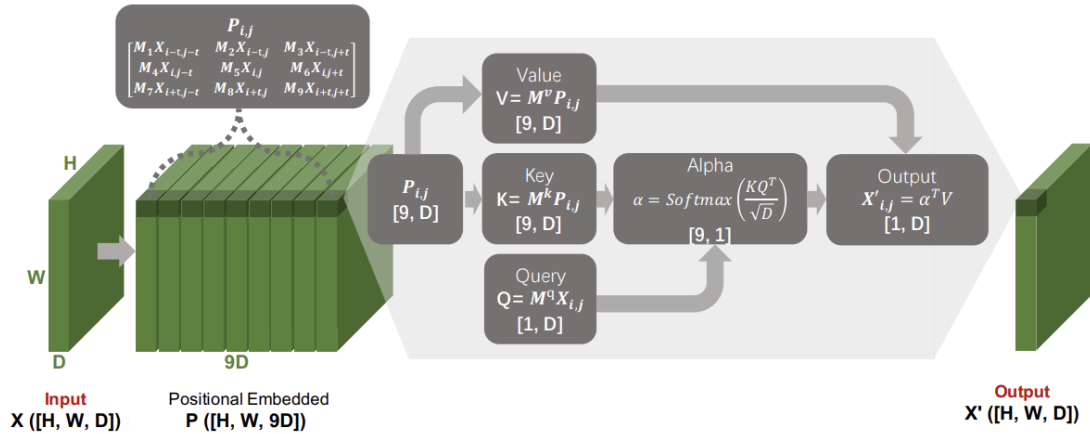


Figure 2.3: Local Self-Attention (LSA) block. For each target pixel $X_{i,j}$, self-attention is computed over its local neighbourhood $P_{i,j}$. Learnable projections M^q , M^k , and M^v generate the Query, Key, and Value representations used in the attention mechanism.

Given a feature map $X \in \mathbb{R}^{H \times W \times D}$, each spatial location $X_{i,j}$ attends to a local $(2N + 1) \times (2N + 1)$ neighbourhood (in this case, $N = 1$, corresponding to a 3×3 neighbourhood) defined using dilation distance t :

$$\begin{bmatrix} X_{i-t,j-t} & X_{i-t,j} & X_{i-t,j+t} \\ X_{i,j-t} & X_{i,j} & X_{i,j+t} \\ X_{i+t,j-t} & X_{i+t,j} & X_{i+t,j+t} \end{bmatrix}$$

The reason for choosing this neighbourhood size will be discussed shortly. For the remainder of this report, we use 3 in place of $(2N + 1)$. The neighbourhood vectors are flattened as $P_{i,j} = [Y_1, Y_2, \dots, Y_9] \in \mathbb{R}^{D \times 9}$, where Y_l denotes the l -th neighbour.

The attention mechanism computes Query, Key, and Value projections:

$$Q = M^q X_{i,j}, \quad K = M^k P_{i,j}, \quad V = M^v P_{i,j}$$

where M^q , M^k , and M^v are learnable projection matrices.

The Local Self-Attention operation is defined as

$$\text{LSA}(X_{i,j}) = \frac{1}{C} \sum_{l=1}^9 \exp \left(\frac{\langle M^k Y_l, M^q X_{i,j} \rangle}{\sqrt{D}} \right) M^v Y_l$$

with normalization constant

$$C = \sum_{l=1}^9 \exp \left(\frac{\langle M^k Y_l, M^q X_{i,j} \rangle}{\sqrt{D}} \right)$$

The Local Self-Attention operation can be interpreted as follows:

- The dot product measures the similarity between the center pixel and its neighbouring features.
- The exponential and subsequent normalization (softmax) produce positive attention weights that sum to one, enabling a probabilistic interpretation.
- The exponential operation also introduces nonlinearity, amplifying highly similar neighbours while suppressing less relevant ones. Consequently, neighbours with greater feature similarity receive higher attention weights than dissimilar neighbours.
- The final output is obtained as a weighted combination of neighbouring features.

Unlike convolution, which applies fixed weighted sums, attention dynamically computes relationships between neighbouring regions based on forensic similarity.

2.2.1 Positional Projection

Standard Transformer architectures use additive positional embeddings:

$$\tilde{X} = X + PE(\text{pos})$$

However, SPAN argues that additive embeddings are unsuitable for forensic feature spaces such as SRM residuals and Bayer inconsistencies, where vector addition does not preserve meaningful structure. Instead, SPAN introduces positional projection matrices M_l , one for each neighbour location. The modified attention equation becomes:

$$\text{LSA}(X_{i,j}) = \frac{1}{C} \sum_{l=1}^9 \exp \left(\frac{\langle M^k M_l Y_l, M^q X_{i,j} \rangle}{\sqrt{D}} \right) M^v Y_l$$

where each learnable matrix M_l injects relative positional information into the corresponding neighbour feature before key projection.

2.2.2 Residual Connection

Each LSA block includes a residual connection:

$$X' = \text{LSA}(X) + X$$

The residual pathway preserves fine-grained forensic information while allowing the attention layers to refine the representation rather than overwrite it entirely.

2.2.3 Pyramid Propagation

The dilation distance t controls how far apart the attended neighbours are from the center pixel while keeping the attention computation fixed to a local 3×3 neighbourhood.

For dilation $t = 1$, attention is computed over the immediate neighbouring pixels:

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} & \mathbf{1} & \mathbf{1} & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} & * & \mathbf{1} & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} & \mathbf{1} & \mathbf{1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

where $*$ denotes the center pixel and the highlighted entries correspond to the attended neighbours. For dilation $t = 3$, the same 3×3 attention pattern is spatially expanded as shown in 2.2.3(a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 & \mathbf{1} & 0 & 0 & \mathbf{1} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 & * & 0 & 0 & \mathbf{1} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 & \mathbf{1} & 0 & 0 & \mathbf{1} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(a) Dilated 3×3 attention with $t = 3$

$$\begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} \\ \mathbf{1} & * & \mathbf{1} & \mathbf{1} & * & \mathbf{1} & \mathbf{1} & * & \mathbf{1} \\ \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} \\ \mathbf{1} & * & \mathbf{1} & \mathbf{1} & * & \mathbf{1} & \mathbf{1} & * & \mathbf{1} \\ \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} \\ \mathbf{1} & * & \mathbf{1} & \mathbf{1} & * & \mathbf{1} & \mathbf{1} & * & \mathbf{1} \\ \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} \end{bmatrix}$$

(b) Effective 9×9 receptive field after two LSA layers

Although attention is still computed over only nine neighbours, each attended location now corresponds to another local 3×3 region from the previous layer. Consequently, information propagates hierarchically across a much larger spatial area.

2.2.3(b) shows how the $t = 3$ layer aggregates information from nine neighbouring 3×3 regions produced by the previous layer.

Thus, after two Local Self-Attention layers, the effective receptive field expands from a local 3×3 neighbourhood to a 9×9 spatial region, corresponding to 81 spatial locations rather than only 49 locations from a naive 7×7 expansion. This hierarchical propagation is enabled through the recursive relationship,

$$X^{(h)} = \text{LSA}(X^{(h-1)}) + X^{(h-1)}$$

where each layer aggregates information from progressively larger spatial regions while preserving the previous representation through residual connections.

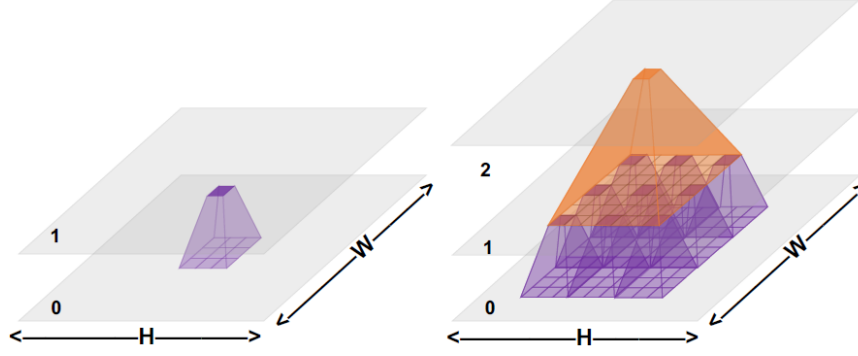


Figure 2.4: Local Attention Pyramid Propagation: Going through self-attention blocks with proper dilation distances (e.g. dilation distance of 1 in the purple blocks and 3 in the orange blocks), information from each pixel location is propagated in a pyramid structure (e.g. each pixel on level-2 encodes information from 9 pixels on level-1 and 81 pixels on level-0).

SPAN stacks five Local Self-Attention layers with dilation distances

$$t \in \{1, 3, 9, 27, 81\}$$

forming a spatial pyramid. As the dilation increases, each pixel representation progressively encodes information from larger image regions. With choice of $N = 1$, the effective spatial coverage grows exponentially:

$$\text{Coverage at layer } h = (2N + 1)^{2h} = 3^{2h}$$

giving receptive fields of:

- Layer 1: 3×3
- Layer 2: 9×9
- Layer 3: 27×27
- Layer 4: 81×81
- Layer 5: 243×243

At higher pyramid levels, each feature vector represents a forensic summary of a large image patch rather than a single pixel. Consequently, self-attention between two feature vectors becomes equivalent to comparing two spatial patches, enabling SPAN to detect inconsistencies between authentic and manipulated regions across large spatial distances while using only local attention operations.

2.2.4 Why 3×3 Blocks?

The neighbourhood (block) size is a hyperparameter in Local Self-Attention. Let

$$M = 2N + 1$$

denote the spatial neighbourhood size of each self-attention block. For an h -layer structure with dilation distances

$$\{1, M, M^2, \dots, M^{h-1}\},$$

each output pixel in the final layer covers an effective receptive field of size

$$M^h \times M^h.$$

Therefore, to ensure coverage of an input image of spatial size $S \times S$, the following condition must hold:

$$M^h \geq S$$

which implies

$$h = O(\log_M S).$$

At each layer, S^2 self-attention computations are required, while each local self-attention block has complexity $O(M^2)$ in both time and memory. Consequently, the total complexity becomes

$$O(S^2 M^2 \log_M S) = O\left(\frac{S^2 M^2 \ln S}{\ln M}\right).$$

To determine the optimal neighbourhood size, we minimize

$$f(M) = \frac{M^2}{\ln M}.$$

Setting the derivative to zero,

$$\frac{d}{dM} f(M) = 0$$

gives

$$\ln M = \frac{1}{2}$$

and therefore

$$M = \sqrt{e} \approx 1.6.$$

Since the neighbourhood size must be an odd integer and 1×1 provides no meaningful neighbourhood interaction, the closest valid choice is

$$M = 3.$$

Thus, 3×3 neighbourhoods provide the best trade-off between computational complexity, receptive-field growth, and multi-scale feature interaction. They maximize spatial coverage under fixed computational and memory budgets while enabling efficient propagation of information across neighbouring regions at multiple scales.

2.3 Decision Module

After pyramid propagation, the attention-enhanced feature representation

$$X^{(5)} \in \mathbb{R}^{H \times W \times 32}$$

is passed through three stacked 1×1 convolutional layers:

$$32 \rightarrow 16 \rightarrow 4 \rightarrow 1$$

The final layer applies a sigmoid activation to produce the prediction map

$$X^{\text{pred}} \in [0, 1]^{H \times W}$$

which represents the pixel-wise probability of manipulation. A binary tampering mask is obtained by thresholding the prediction map at a threshold τ (default $\tau = 0.5$), where pixels with values greater than τ are classified as manipulated and the remaining pixels as authentic. Because all convolutions are 1×1 , no spatial smoothing occurs during prediction, preserving sharp manipulation boundaries.

2.4 Training Objective

SPAN is trained using Binary Cross-Entropy (BCE) loss:

$$\mathcal{L} = -\frac{1}{HW} \sum_{i,j} \left[B_{i,j} \log X_{i,j}^{\text{pred}} + (1 - B_{i,j}) \log(1 - X_{i,j}^{\text{pred}}) \right]$$

where $B \in \{0, 1\}^{H \times W}$ is the ground-truth manipulation mask.

The BCE objective is appropriate because forgery localization is a pixel-wise binary classification problem. The normalization factor $\frac{1}{HW}$ ensures that the loss remains independent of image size and stabilizes gradient magnitudes during optimization.

Experiments & Performance Analysis

3.1 Experimental Setup

All experiment scripts, evaluation pipelines, and robustness analysis code used in this study are publicly available in [GitHub repository](#).

3.1.1 Datasets

Three publicly available benchmark datasets were used for evaluation. Crucially, no fine-tuning was performed on any dataset — pre-trained weights were loaded directly from the official repositories, making all results a test of genuine zero-shot generalisation ability.

Columbia Uncompressed Image Splicing Dataset

The Columbia dataset is one of the earliest and most widely used benchmarks for image splicing detection. It contains 180 images — 32 authentic and 148 tampered — created by cutting regions from one photograph and pasting them into another, all sourced from uncompressed TIFF images. Since the images are uncompressed, forensic traces such as camera sensor noise and demosaicing artifacts remain well preserved. Each forged image is accompanied by a pixel-level binary ground truth mask precisely localising the manipulated region.

This dataset therefore acts as a near best-case benchmark for forensic localisation: if a model cannot detect manipulations here, it is fundamentally incapable of exploiting forensic traces. **Evaluation metric:** Pixel-level AUC (Area Under the ROC Curve).

Coverage Dataset

The Coverage dataset evaluates copy-move forgery detection, where a region within the same image is copied and pasted elsewhere. It contains 100 manipulated images, each paired with a pixel-level grayscale mask.

Unlike splicing, copy-move manipulation does not introduce cross-image noise inconsistencies because both regions originate from the same camera source. Detection is therefore substantially harder, requiring the model to identify duplicated structures and local statistical inconsistencies rather than simple sensor-noise discontinuities.

This dataset is consequently a stronger test of SPAN’s spatial attention mechanism and its ability to compare neighbouring image regions across scales. **Evaluation metric:** Pixel-level AUC and F1-score.

CASIA v1 Dataset

CASIA v1 is a mixed-manipulation dataset containing 921 forged images and 800 authentic images spanning both splicing and copy-move forgeries. Unlike Columbia and Coverage, CASIA v1 does not provide pixel-level masks — only image-level real/fake labels are available.

Consequently, evaluation must be performed at the image level. The maximum predicted value in the SPAN heatmap is used as the image-level manipulation confidence score. This provides a weaker evaluation signal because localisation quality is not measured directly.

CASIA v1 therefore evaluates image-level detection capability rather than precise manipulation localisation. **Evaluation metric:** Image-level AUC and F1-score.

3.1.2 Models Under Evaluation

Both SPAN (Spatial Pyramid Attention Network) and ManTraNet were evaluated. Pre-trained weights were obtained from their respective official repositories.

The original implementations were provided in TensorFlow using `.h5` checkpoints. Due to incompatibilities with modern TensorFlow versions, both architectures were re-implemented in PyTorch. Model parameters were transferred manually using `h5py`-based weight conversion by mapping TensorFlow layer names to corresponding PyTorch tensors. GitHub Copilot assistance was used for boilerplate code generation and code conversion. No layers were retrained or fine-tuned; only weight transfer was performed.

The original papers report results using FP32 precision. Due to GPU memory limitations and computational constraints, all experiments in this work were executed using FP16 mixed precision. While this may introduce minor numerical variation — particularly in attention softmax computations — it does not alter the overall qualitative trends.

3.1.3 Robustness Perturbation Conditions

To evaluate robustness, each test image was subjected to controlled perturbations before inference. These perturbations simulate realistic post-processing operations commonly applied to conceal manipulation traces. The details of the applied perturbations are summarized in Table 3.1.

Forensic detectors such as SPAN and ManTraNet rely on subtle statistical inconsistencies — noise patterns, demosaicing traces, and boundary artifacts — introduced during image manipulation. Operations such as blur, aggressive rescaling, or heavy JPEG compression homogenise these signals and therefore reduce detection performance.

Condition	Description	Effect on Forensic Traces
No manipulation	Raw image as-is	Baseline condition; forensic traces fully preserved
Resize $0.78\times$	Mild downscaling and upscaling	Slight interpolation smooths high-frequency noise patterns
Resize $0.25\times$	Aggressive downscaling and upscaling	Destroys fine-grained sensor noise and smears DCT block boundaries
Blur ($k = 3$)	Mild Gaussian blur	Partially suppresses local texture and noise fingerprints
Blur ($k = 15$)	Heavy Gaussian blur	Strong low-pass filtering removes camera noise artifacts
Gaussian Noise $\sigma = 3$	Mild additive noise	Partially masks manipulation boundaries while preserving signal
Gaussian Noise $\sigma = 15$	Heavy additive noise	Synthetic noise overwhelms the original forensic fingerprint
JPEG $q = 100$	Near-lossless JPEG compression	Introduces JPEG structure with minimal quantisation loss
JPEG $q = 50$	Medium-quality JPEG compression	Distorts DCT coefficients and introduces blocking artifacts

Table 3.1: Robustness perturbation conditions applied during evaluation.

3.1.4 Evaluation Protocol

- Columbia and Coverage — Pixel-Level Evaluation:** Both SPAN and ManTraNet produce a heatmap of the same spatial resolution as the input image, where each pixel value represents the model’s predicted probability that the pixel has been manipulated. This heatmap is compared directly against the ground-truth binary mask pixel-by-pixel. AUC (Area Under the ROC Curve) is computed by varying the decision threshold from 0 to 1 and measuring the trade-off between true positive rate and false positive rate across all pixels in the dataset — a threshold-agnostic measure of overall ranking ability. F1-score is computed at the single optimal threshold that maximises the harmonic mean of precision and recall, providing a threshold-sensitive measure of localisation quality.
- CASIA v1 — Image-Level Evaluation:** CASIA v1 does not provide pixel-level ground-truth masks — only binary image-level labels (authentic or forged) are available. It is therefore impossible to evaluate localisation quality directly. Instead, a single scalar confidence score must be derived from the predicted heatmap for each image and compared against its label.

The maximum value of the predicted heatmap is used as this image-level score: if the model assigns high manipulation probability to *any* pixel in the image, that image is considered likely forged. This is a natural choice because a localisation model does not predict a global image-level score by design — it predicts per-pixel scores. Taking the maximum is the most faithful way to convert a spatial prediction into

an image-level decision without discarding the model’s strongest evidence. AUC and F1 are then computed over these image-level scores against the binary labels.

This evaluation is inherently weaker than pixel-level evaluation: a model may correctly localise the tampered region but produce a diffuse, low-confidence heatmap, causing it to score poorly despite being qualitatively correct. CASIA v1 therefore measures *detection* ability rather than *localisation* precision.

- For all datasets, perturbations were applied only to the input image before inference. Ground-truth masks, wherever applicable, remained unchanged.

3.2 Results

3.2.1 Columbia Dataset — Pixel-Level AUC

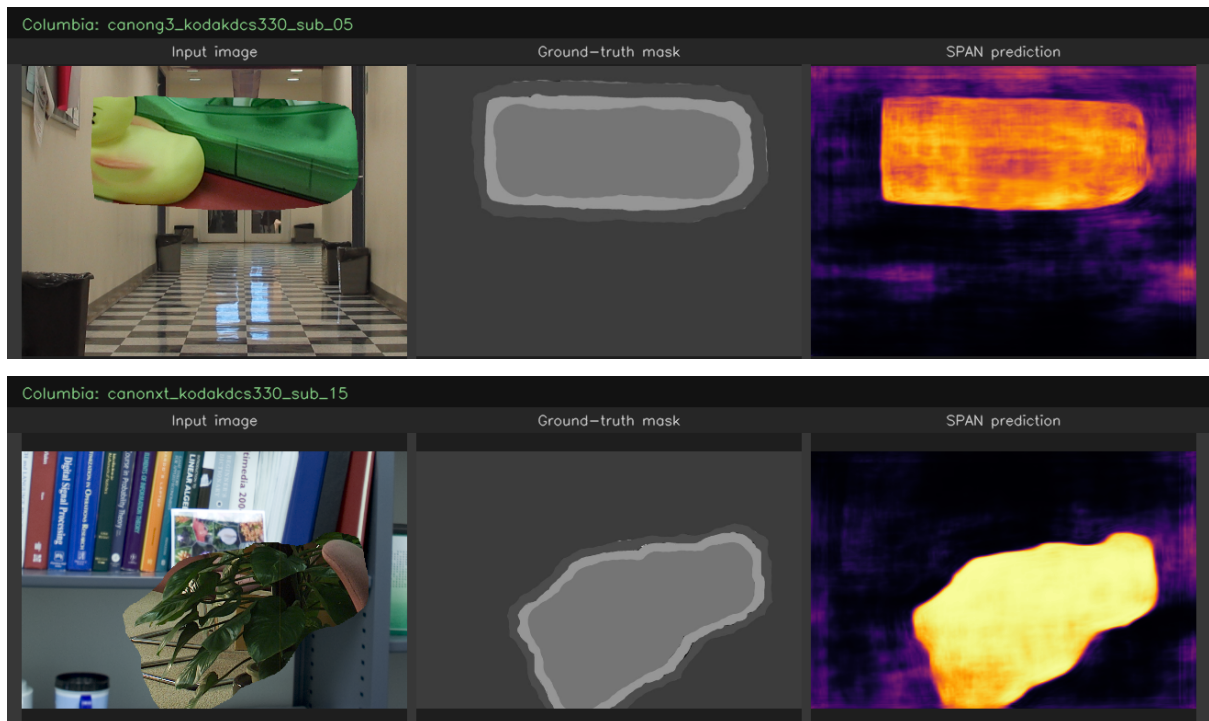


Figure 3.1: Sample results on the Columbia dataset. Left: manipulated image, middle: ground-truth mask, right: SPAN prediction heatmap.

The Columbia dataset serves as the primary benchmark for manipulation localisation due to its clean, uncompressed splicing masks. SPAN consistently outperforms ManTraNet across all perturbation conditions. The advantage is especially large under JPEG $q = 100$, where the underlying forensic traces remain nearly intact.

Performance degrades under aggressive blur and resize operations because these operations destroy high-frequency forensic information on which both models depend.

Condition	SPAN	ManTraNet	Δ
No manipulation	92.86	72.09	+20.77
Resize $0.78\times$	84.30	62.30	+22.00
Resize $0.25\times$	65.12	61.03	+4.09
Blur ($k = 3$)	66.95	62.57	+4.38
Blur ($k = 15$)	66.07	63.77	+2.30
Noise $\sigma = 3$	73.21	59.62	+13.59
Noise $\sigma = 15$	67.84	51.94	+15.90
JPEG $q = 100$	91.91	66.35	+25.56
JPEG $q = 50$	71.03	57.70	+13.33

Table 3.2: Pixel-level AUC on the Columbia dataset.

3.2.2 Coverage Dataset — Pixel-Level AUC and F1

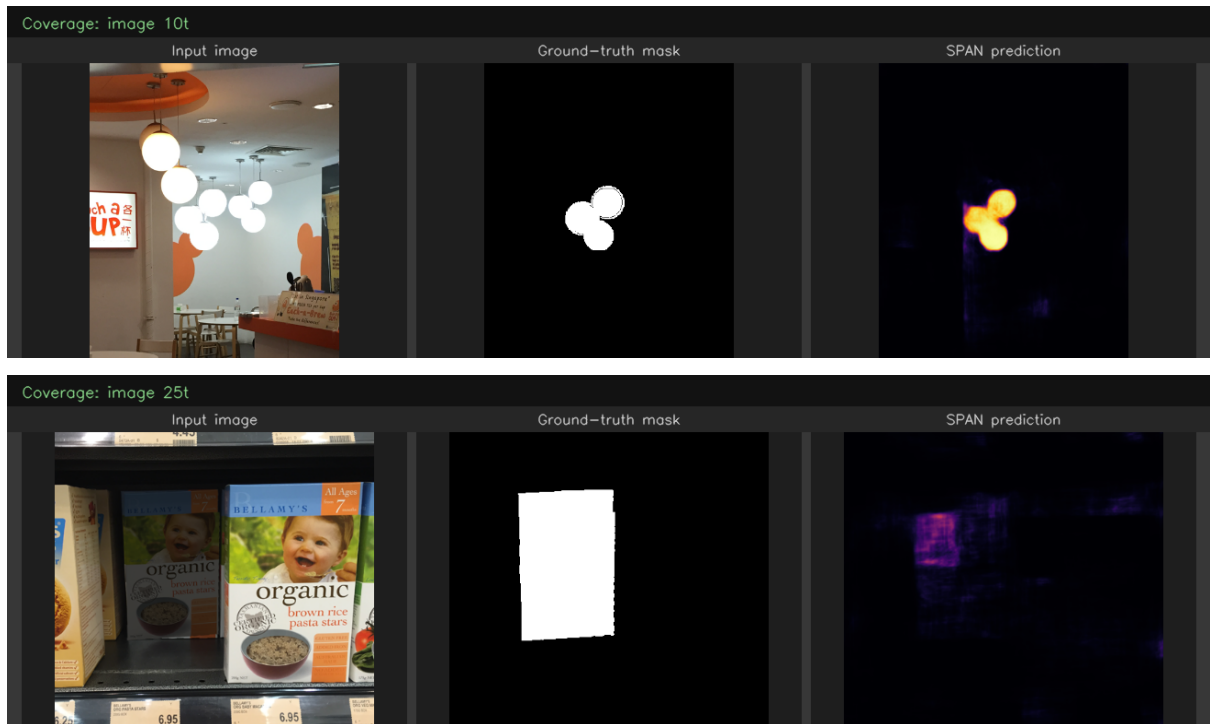


Figure 3.2: Sample results on the Coverage dataset. Left: manipulated image, middle: ground-truth mask, right: SPAN prediction heatmap.

SPAN again achieves consistently higher AUC values across all perturbations, indicating that the pyramid attention mechanism generalises effectively to copy-move manipulation despite the absence of cross-image noise inconsistencies.

However, F1 scores remain relatively low for both models because copy-move boundaries are visually and statistically more subtle than splicing boundaries.

Condition	SPAN AUC	MT AUC	SPAN F1	MT F1
No manipulation	91.37	71.61	44.95	24.31
Resize 0.78×	74.05	65.16	21.85	21.71
Resize 0.25×	63.13	62.61	19.57	20.60
Blur ($k = 3$)	65.06	64.45	19.74	22.38
Blur ($k = 15$)	63.36	61.67	19.57	22.19
Noise $\sigma = 3$	69.03	60.19	19.99	20.10
Noise $\sigma = 15$	60.26	53.59	20.52	19.41
JPEG $q = 100$	87.63	66.14	35.35	21.74
JPEG $q = 50$	65.99	59.49	19.57	19.99

Table 3.3: Pixel-level evaluation on the Coverage dataset.

3.2.3 CASIA v1 Dataset — Image-Level Evaluation

The CASIA results demonstrate the weakness of image-level evaluation for localisation-oriented architectures. Although SPAN substantially outperforms ManTraNet in AUC, absolute performance is lower than Columbia and Coverage because localisation quality is not evaluated directly.

ManTraNet exhibits nearly constant F1 across all perturbations, strongly suggesting degenerate prediction behaviour on this dataset.

Condition	SPAN AUC	MT AUC	SPAN F1	MT F1
No manipulation	73.11	47.96	63.14	69.72
Resize 0.78×	59.99	45.26	35.15	69.72
Resize 0.25×	48.67	45.79	36.86	69.72
Blur ($k = 3$)	53.79	46.62	10.50	69.72
Blur ($k = 15$)	49.18	47.13	6.81	69.72
Noise $\sigma = 3$	48.72	49.44	67.21	69.72
Noise $\sigma = 15$	52.00	57.35	69.65	69.72
JPEG $q = 100$	72.54	48.03	59.54	69.72
JPEG $q = 50$	60.46	41.76	44.27	69.67

Table 3.4: Image-level evaluation on the CASIA v1 dataset.

3.3 Overall Analysis and Discussion

3.3.1 Robustness Trends

Across all datasets, the most robust conditions are the unperturbed setting and JPEG $q = 100$, where forensic traces remain largely preserved.

Moderate degradation occurs under mild Gaussian noise and mild resize operations, while aggressive blur, strong resize, and heavy additive noise substantially reduce performance by erasing high-frequency forensic information.

3.3.2 SPAN vs. ManTraNet

SPAN consistently outperforms ManTraNet because it explicitly models spatial consistency across multiple scales using pyramid self-attention. ManTraNet primarily focuses on anomaly-map extraction from forensic traces, whereas SPAN additionally compares neighbouring regions hierarchically using dilated local attention. This enables SPAN to localise manipulation boundaries more accurately across both splicing and copy-move scenarios.

3.3.3 Limitations and Resource Constraints

Both SPAN and ManTraNet fundamentally depend on the preservation of high-frequency forensic traces. Consequently, image degradations such as blur, aggressive resizing, JPEG compression, and noise significantly reduce localisation performance, as demonstrated in the robustness analysis. This degradation becomes especially severe when manipulation artifacts are weak or heavily post-processed.

Slight quantitative deviations from the values reported in the original papers were observed, likely due to the use of FP16 precision instead of FP32. While FP16 enabled feasible execution under hardware constraints, its reduced numerical precision may introduce small variations in attention computations, particularly on challenging datasets with weak forensic signals.

Due to hardware and runtime constraints, experiments were limited to zero-shot evaluation on Columbia, Coverage, and CASIA v1 datasets using publicly released pretrained weights without additional fine-tuning. Larger-scale datasets such as CASIA v2 and more computationally intensive adaptation experiments could not be explored within the available resources.

Similarly, several architectural variants and ablation configurations reported in the original SPAN paper—such as replacing the residual fusion of multi-scale attention features with convolutional LSTM-based aggregation, and substituting sinusoidal position embeddings (as used in the Transformer architecture) with position projection—were not reproduced due to their substantially higher training and evaluation cost. Comparisons with additional forgery localisation approaches such as H-LSTM and RGB-N were also beyond the scope of this study.

Another limitation arises from dataset annotations. CASIA v1 does not provide pixel-level ground-truth masks, making it unsuitable for evaluating precise localisation capability and restricting evaluation primarily to image-level detection.

3.4 Conclusion

This work evaluated SPAN and ManTraNet as zero-shot detectors across three benchmark datasets under multiple perturbation conditions. SPAN consistently outperformed ManTraNet on both splicing and copy-move localisation, demonstrating that hierarchical pyramid self-attention provides substantially stronger spatial reasoning than anomaly-map-based approaches. Both models share a fundamental limitation: dependence on forensic noise fingerprints that are easily destroyed by post-processing such as blur and aggressive rescaling.

References

- **SPAN**: Spatial Pyramid Attention Network for Image Manipulation Localization (2020) — Original paper introducing the SPAN architecture. ([Link](#))
- **ManTra-Net**: Manipulation Tracing Network for Detection and Localization of Image Forgeries With Anomalous Features (2019) — Original paper proposing ManTra-Net. ([Link](#))
- **Attention Is All You Need** (2017) — Vaswani et al., original paper introducing the Transformer architecture. ([Link](#))
- **SPAN — Official Implementation Repository**. ([Link](#))
- **ManTra-Net — Official Implementation Repository**. ([Link](#))
- **Columbia Image Splicing Detection Dataset** — Splicing-only dataset containing high-resolution manipulated images with edge-based ground-truth masks. ([Link](#))
- **Coverage Dataset** — Copy-move forgery dataset containing realistic object-level manipulations and post-processing operations. ([Link](#))
- **CASIA v1.0 / v2.0 Dataset** — Large-scale forgery dataset containing splicing and copy-move manipulations with additional post-processing effects such as blur and filtering. ([Link](#))